RAG Learning

Required Practicals (RP1–RP8)

AQA GCSE Chemistry (8462) — separate Chemistry

Practical	Spec reference	Paper
RP1 — Making Salts	4.4.2.3	Paper 1
RP2 — Neutralisation (Titration) (chemistry only)	4.4.2.5	Paper 1
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RP1 — Making Salts

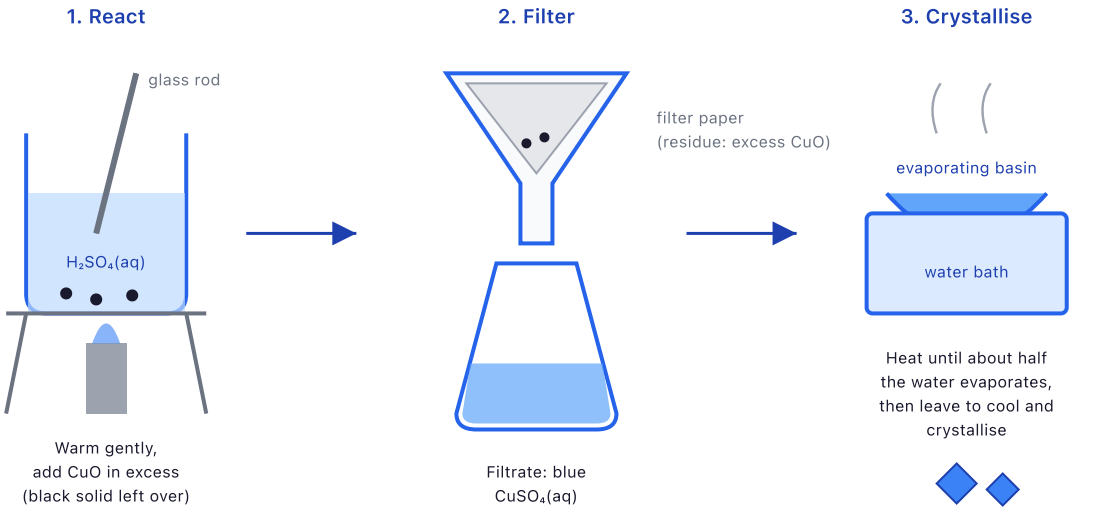
> **Aim:** Prepare a pure, dry sample of a soluble salt from an insoluble oxide or carbonate, using a Bunsen burner to heat dilute acid and a water bath or electric heater to evaporate the solution. The standard example is making copper sulfate crystals from copper(II) oxide and dilute sulfuric acid.

Apparatus

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- Dilute sulfuric acid (1.0 mol/dm³) and copper(II) oxide powder (or copper carbonate)
- Measuring cylinder (e.g. 50 cm³)
- Beaker (100 cm³), glass rod and spatula
- Bunsen burner, tripod, gauze and heatproof mat
- Filter funnel, filter paper and conical flask
- Evaporating basin and water bath (or electric heater)
- Crystallising dish, paper towels
- Eye protection

RP1: Making Copper Sulfate Crystals — Three Stages



$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

Method

1. Measure **40 cm³** of dilute sulfuric acid with a measuring cylinder and pour it into a beaker.
2. Warm the acid gently using a Bunsen burner (do not boil). Warming speeds up the reaction.
3. Add copper(II) oxide a spatula at a time, stirring with a glass rod, until no more dissolves and some black solid remains — the copper(II) oxide is now **in excess**, so all the acid has reacted.
4. Allow to cool slightly, then **filter** the mixture into a conical flask to remove the excess copper(II) oxide. The filtrate is blue copper sulfate solution.
5. Pour the filtrate into an evaporating basin and heat it over a **water bath** (or electric heater) to evaporate about half the water, concentrating the solution.

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6. Pour the hot concentrated solution into a crystallising dish and leave it in a cool place for at least 24 hours for crystals to form.
7. Remove the crystals, pat them dry between paper towels (or leave to dry in a warm place).

Key Measurements and Calculations

- **Equation:** $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$
- With a **carbonate** (e.g. $CuCO_3$), effervescence (CO_2) is seen; it is in excess when fizzing stops and solid remains.
- **Product:** blue crystals of hydrated copper sulfate, $CuSO_4 \cdot 5H_2O$
- **Percentage yield** (if mass measured) = (actual mass ÷ theoretical mass) × 100
- **Choosing the acid:** hydrochloric acid → chlorides; sulfuric acid → sulfates; nitric acid → nitrates.

> **Exam Tip:** Explain *why* each step is done: excess base ensures all the acid is used up (so the salt is not contaminated with acid); filtering removes the unreacted solid; a water bath heats gently so the salt does not decompose or spit.

Sources of Error & Minimisation

- **Unreacted acid remaining:** Always add the base until it is clearly in excess (solid remains after stirring).
- **Loss of product:** Some solution is left on the filter paper and in the apparatus; rinse with a little distilled water. This lowers the percentage yield.
- **Small or powdery crystals:** Evaporating too far or cooling too quickly gives small crystals. Evaporate only until crystals start to form at the edge, then cool slowly.
- **Crystals not fully dry:** Dry thoroughly between paper towels or in a warm oven (not too hot) before weighing.

Safety Considerations

- **Copper(II) oxide and copper sulfate:** harmful if swallowed and irritant to eyes; copper sulfate is dangerous to the environment. Wear eye protection and do not pour down the sink.

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- **Dilute sulfuric acid:** irritant; wear eye protection and wipe up spills.
- **Heating:** do not boil the acid or evaporate to dryness — hot solution can spit. Allow apparatus to cool before handling.

RP2 — Neutralisation: Titration (chemistry only)

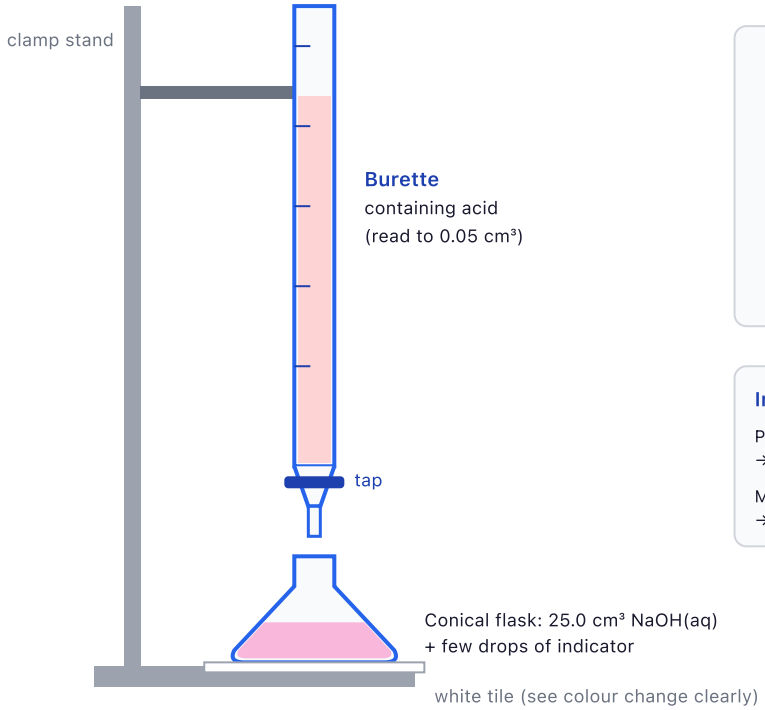
> **Aim:** Determine the reacting volumes of solutions of a strong acid and a strong alkali by titration, and (Higher) use these volumes to find the concentration of one solution when the other is known.

Apparatus

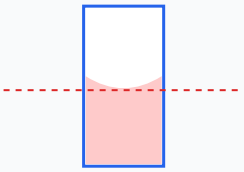
- Burette (50 cm^3), clamp stand and white tile
- Volumetric pipette (25.0 cm^3) with pipette filler
- Conical flask (250 cm^3)
- Small funnel (for filling the burette)
- Dilute hydrochloric acid (or sulfuric acid) and sodium hydroxide solution
- Indicator: phenolphthalein or methyl orange (a single-colour-change indicator, not universal indicator)
- Eye protection

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RP2: Titration Apparatus



Reading the burette



Read the bottom of the meniscus at eye level

Indicator colours

Phenolphthalein: pink (alkali) → colourless (acid)

Methyl orange: yellow (alkali) → red (acid)

Method

1. Use a pipette and pipette filler to transfer exactly **25.0 cm³** of sodium hydroxide solution into a clean conical flask.
2. Add a few drops of indicator (e.g. phenolphthalein — the solution turns pink) and place the flask on a white tile under the burette.
3. Using a funnel, fill the burette with acid. Remove the funnel, run a little acid through the tap to fill the jet, and record the **initial burette reading** to 2 decimal places (to the nearest 0.05 cm³).
4. Add the acid quickly at first while swirling the flask, to find the approximate end point — this is the **rough titration**.
5. Repeat with a fresh 25.0 cm³ of alkali. This time, add the acid quickly until about 2 cm³ before the rough end point, then **drop by drop**, swirling constantly, until the indicator **just** changes colour permanently (the end point).
6. Record the final burette reading and calculate the **titre** (volume of acid added) = final reading – initial reading.
7. Repeat until you have at least two **concordant** titres (within 0.10 cm³ of each other), and calculate the mean of the concordant titres.

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Key Measurements and Calculations

- **Titre:** final burette reading – initial burette reading (cm^3)
- **Mean titre:** average of concordant results only (ignore the rough titre and anomalies)
- **Moles:** n = concentration (mol/dm^3) × volume (dm^3); convert cm^3 to dm^3 by ÷ 1000
- **Worked example (Higher):** 25.0 cm^3 of NaOH is neutralised by a mean titre of 20.00 cm^3 of 0.100 mol/dm^3 HCl.
  Moles HCl = 0.100 × 0.02000 = 0.00200 mol
  HCl + NaOH → NaCl + H_2O (1 : 1), so moles NaOH = 0.00200 mol
  Concentration NaOH = 0.00200 ÷ 0.0250 = **0.0800 mol/dm^3**
- **Sulfuric acid:** $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$, so moles NaOH = 2 × moles H_2SO_4.
- **Concentration in g/dm^3** = concentration in mol/dm^3 × M_r

> **Exam Tip:** Universal indicator is not used in titrations because it changes colour gradually through many colours, so a sharp end point cannot be seen. Always use a single-colour-change indicator such as phenolphthalein or methyl orange.

Sources of Error & Minimisation

- **Overshooting the end point:** Add acid drop by drop near the end point and swirl continuously.
- **Parallax error:** Read the bottom of the meniscus at eye level.
- **Air bubble in the burette jet:** Run some acid through the tap before taking the initial reading.
- **Leaving the funnel in the burette:** Drips from the funnel change the reading — remove it before titrating.
- **Contaminated apparatus:** Rinse the burette with the acid and the pipette with the alkali before use; rinse the conical flask with distilled water only.
- **Random error:** Repeat until concordant results are obtained, and calculate the mean.

Safety Considerations

- **Sodium hydroxide:** irritant/corrosive, especially to eyes. Wear eye protection at all times.
- **Acids:** irritant; wash spills off skin with plenty of water.

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- **Filling the burette:** fill below eye level (e.g. stand the burette on the bench or lower it) to avoid splashes to the face.
- **Pipette:** always use a pipette filler — never pipette by mouth.

RP3 — Electrolysis

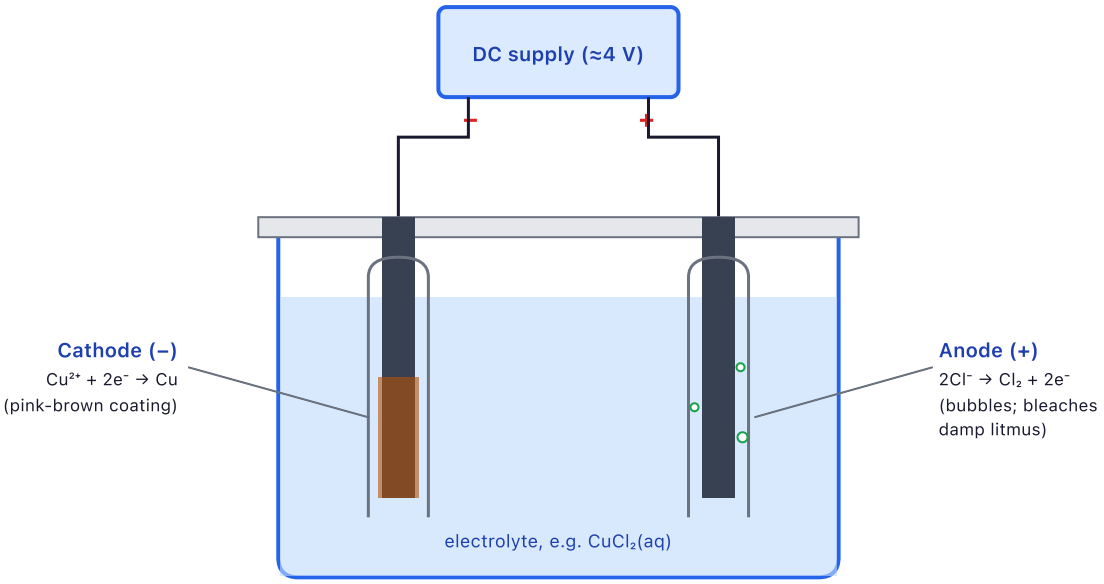
> **Aim:** Investigate what happens when aqueous solutions are electrolysed using inert electrodes, identifying the products at each electrode (e.g. copper(II) chloride, sodium chloride, sodium sulfate and copper(II) sulfate solutions).

Apparatus

- Low-voltage DC power supply (set to about 4 V)
- Two inert carbon (graphite) electrodes in a lid or electrode holder
- Beaker (100 cm^3) or petri dish
- Connecting leads with crocodile clips
- Two small test tubes (to collect gases) filled with the solution
- Blue litmus paper, wooden splints, Bunsen burner
- Solutions to test: copper(II) chloride, sodium chloride, sodium sulfate, copper(II) sulfate (all about 0.5 mol/dm^3)
- Eye protection

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RP3: Electrolysis of an Aqueous Solution



Inert graphite electrodes; small test tubes of electrolyte collect any gas

Method

1. Pour about 50 cm³ of copper(II) chloride solution into the beaker.
2. Fit the lid holding the two carbon electrodes, making sure the electrodes do not touch each other.
3. Connect the electrodes to the DC power supply with leads and crocodile clips: the electrode connected to the negative terminal is the **cathode**, and the one connected to the positive terminal is the **anode**.
4. Fill two small test tubes with the solution and invert one over each electrode to collect any gas.
5. Switch on at about 4 V and observe what happens at each electrode for a few minutes.
6. Test any gas: hold **damp blue litmus paper** near the anode (chlorine bleaches it); test collected gas with a **glowing splint** (oxygen relights it) or a **lighted splint** (hydrogen gives a squeaky pop).
7. Switch off, then remove and inspect the cathode for any metal coating.
8. Clean the electrodes and repeat with the other solutions (sodium chloride, sodium sulfate, copper(II) sulfate).

Key Measurements and Calculations — Expected Results

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Solution	Cathode (–)	Anode (+)
Copper(II) chloride	Copper (pink-brown coating)	Chlorine (bleaches damp litmus)
Sodium chloride	Hydrogen (squeaky pop)	Chlorine
Sodium sulfate	Hydrogen	Oxygen (relights glowing splint)
Copper(II) sulfate	Copper	Oxygen

- **Cathode rule:** the metal is produced if it is less reactive than hydrogen; otherwise hydrogen is produced.
- **Anode rule:** oxygen is produced unless halide ions are present, in which case the halogen is produced.
- **Half equations (Higher):** $2H^+ + 2e^- \rightarrow H_2$ (cathode); $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ (anode). Reduction occurs at the cathode, oxidation at the anode.
- A quantitative extension: weigh the cathode before and after to find the mass of copper deposited, and investigate how it depends on current or time.

Sources of Error & Minimisation

- **Electrodes touching:** causes a short circuit — keep them apart in a fixed holder.
- **Gas escaping before testing:** collect gas in an inverted test tube over the electrode rather than testing in the open.
- **Contamination between solutions:** clean and dry the electrodes and beaker between each solution.
- **Copper falling off the cathode (if weighing):** use a low current, and rinse with distilled water then propanone and dry carefully before reweighing.

Safety Considerations

- **Chlorine gas:** toxic — run the electrolysis for only a short time, in a well-ventilated room; anyone with asthma should take extra care.
- **Copper(II) chloride/sulfate:** harmful and irritant; wear eye protection and do not pour copper solutions down the sink.

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- **Electrical:** use low voltage only; switch off before handling the electrodes.
- **Hydrogen:** flammable — only test small volumes away from other flames.

RP4 — Temperature Changes

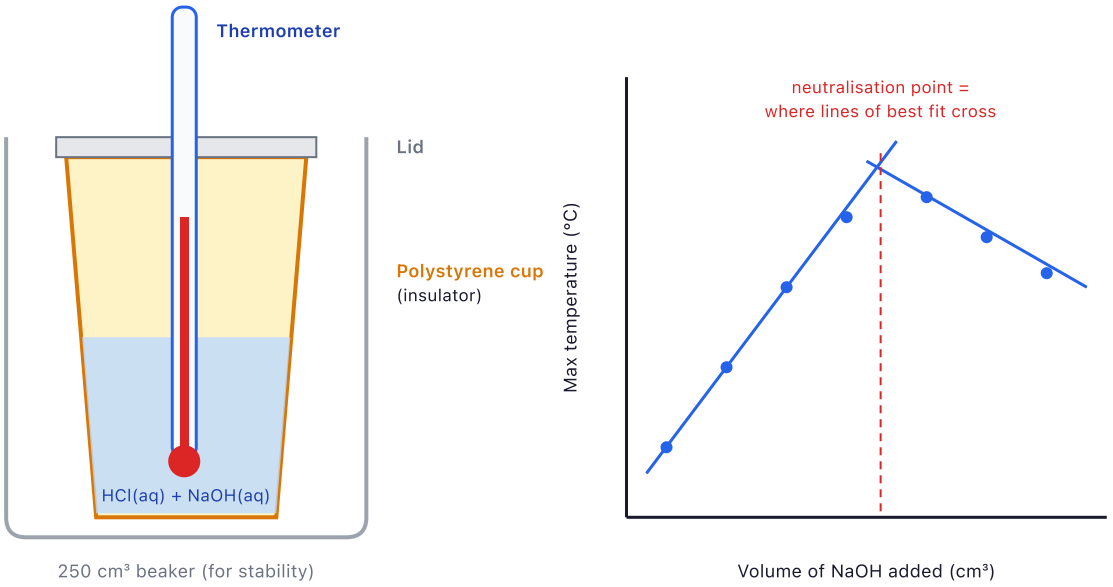
> **Aim:** Investigate the variables that affect temperature changes in reacting solutions, such as acid plus metals, acid plus carbonates, neutralisations, or displacement of metals. The standard example measures the temperature change when different volumes of sodium hydroxide are added to hydrochloric acid.

Apparatus

- Polystyrene cup with a lid (with a hole for the thermometer), standing in a 250 cm^3 beaker for stability
- Thermometer (–10 to 110 °C) or temperature probe
- Measuring cylinders (10 cm^3 and 50 cm^3)
- Dilute hydrochloric acid and sodium hydroxide solution (both about 2 mol/dm^3)
- Glass rod (for stirring)
- Eye protection

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RP4: Temperature Change Apparatus and Typical Results



Method

1. Measure **30 cm³** of dilute hydrochloric acid with a measuring cylinder and pour it into the polystyrene cup, standing in a beaker.
2. Record the **starting temperature** of the acid.
3. Measure **5 cm³** of sodium hydroxide solution and add it to the cup. Fit the lid, stir with the thermometer or glass rod, and record the **highest temperature** reached.
4. Add a further 5 cm³ of sodium hydroxide, stir, and record the highest temperature again.
5. Continue adding 5 cm³ portions until a total of 40 cm³ has been added.
6. Repeat the whole experiment and calculate the mean maximum temperature for each volume.
7. Plot a graph of maximum temperature (y-axis) against volume of sodium hydroxide added (x-axis). Draw two straight lines of best fit — one through the rising points and one through the falling points — and find where they cross.

Key Measurements and Calculations

- **Independent variable:** volume of sodium hydroxide added (cm³)
- **Dependent variable:** highest temperature reached (°C)

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- **Control variables:** volume and concentration of acid, concentration of alkali, type of cup, starting temperature
- **Temperature change:** ΔT = highest temperature – starting temperature
- **Interpretation:** the temperature rises as more acid is neutralised (exothermic); the peak is where the acid is exactly neutralised; after that, extra cooler alkali absorbs energy and dilutes the mixture, so the temperature falls.
- A rise in temperature = **exothermic**; a fall in temperature (e.g. citric acid + sodium hydrogencarbonate) = **endothermic**.

Sources of Error & Minimisation

- **Energy transfer to or from the surroundings:** use a polystyrene cup (good insulator) with a lid; stand it in a beaker so it does not tip.
- **Missing the maximum temperature:** stir continuously and watch the thermometer until it stops rising.
- **Uneven temperature in the mixture:** stir thoroughly after each addition.
- **Thermometer resolution:** a digital probe gives a higher resolution (0.1 °C) than a liquid-in-glass thermometer.
- **Random error:** repeat each set of readings and calculate a mean; identify anomalies.

Safety Considerations

- **Sodium hydroxide:** irritant/corrosive to eyes — wear eye protection.
- **Hydrochloric acid:** irritant; wipe up spills immediately.
- **Glass thermometer:** do not stir vigorously with a glass thermometer as it may break; use a glass rod if possible.

RP5 — Rates of Reaction

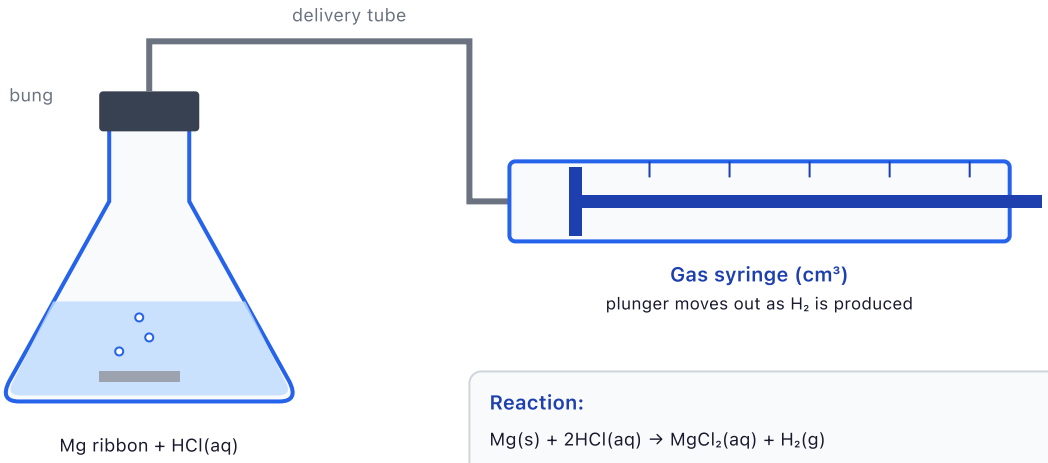
> **Aim:** Investigate how changes in concentration affect the rates of reactions by (a) a method involving measuring the volume of a gas produced, and (b) a method involving a change in colour or turbidity (the "disappearing cross").

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Apparatus — Method A: Volume of Gas

- Conical flask with a single-holed bung and delivery tube
- Gas syringe (or an upturned measuring cylinder in a trough of water)
- Magnesium ribbon (3 cm lengths) and hydrochloric acid at several concentrations (e.g. 0.5, 1.0, 1.5, 2.0 mol/dm³)
- Measuring cylinder (50 cm³), stopwatch, clamp stand
- Eye protection

RP5 Method A: Measuring the Volume of Gas Produced



Method A — Volume of Gas

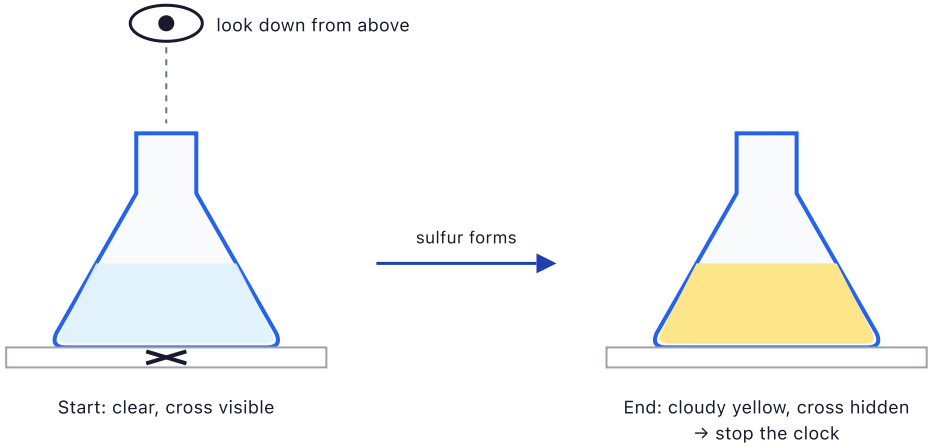
1. Set up the conical flask, bung, delivery tube and gas syringe. Check the syringe plunger is pushed fully in (reads 0 cm³).
2. Measure **50 cm³** of 0.5 mol/dm³ hydrochloric acid into the conical flask.
3. Add a **3 cm** strip of magnesium ribbon, immediately replace the bung and start the stopwatch.
4. Record the volume of gas in the syringe every **10 seconds** until no more gas is produced.
5. Repeat with each of the other acid concentrations, keeping the volume of acid and length of magnesium the same.
6. Plot a graph of volume of gas (y-axis) against time (x-axis) for each concentration on the same axes.

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Apparatus — Method B: Disappearing Cross (Turbidity)

- Conical flask (100 cm^3) and a sheet of paper marked with a black cross
- Sodium thiosulfate solution at several concentrations (e.g. 10, 20, 30, 40 g/dm^3)
- Dilute hydrochloric acid (1.0 mol/dm^3)
- Measuring cylinders (10 cm^3 and 50 cm^3), stopwatch
- Eye protection

RP5 Method B: The Disappearing Cross



$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$

Method B — Disappearing Cross

1. Measure **10 cm^3** of sodium thiosulfate solution into the conical flask (or a fixed volume of thiosulfate made up to a fixed total volume with water).
2. Place the flask on the black cross.
3. Measure **10 cm^3** of dilute hydrochloric acid, add it to the flask, swirl and start the stopwatch.
4. Look down through the flask from above and stop the clock when the cross can **no longer be seen**.
5. Repeat for each concentration of sodium thiosulfate, keeping the total volume, acid concentration and the same person judging the end point.
6. Calculate the rate as `1 / time` (s^{-1}) and plot rate against concentration.

Key Measurements and Calculations

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- **Independent variable:** concentration of acid (Method A) or sodium thiosulfate (Method B)
- **Dependent variable:** volume of gas at each time (A); time for the cross to disappear (B)
- **Control variables:** temperature, volumes of solutions, mass/length and surface area of magnesium, same cross and observer
- **Mean rate** = quantity of product formed ÷ time taken (e.g. cm^3/s)
- **Rate at a given time (Higher):** draw a tangent to the curve and calculate its gradient
- **Method B:** rate $\propto$ 1/time; higher concentration → shorter time → faster rate
- **Explanation:** a higher concentration means more reactant particles in the same volume, so collisions are more frequent and the rate increases.

> **Exam Tip:** On a volume–time graph, a steeper initial gradient means a faster rate. If the acid is in excess and the same mass of magnesium is used each time, every curve levels off at the same final volume of gas.

Sources of Error & Minimisation

- **Gas escaping before the bung is replaced:** insert the bung immediately; alternatively place the magnesium in a small tube inside the flask and tip it over once sealed.
- **Gas syringe sticking:** check that the plunger moves freely before starting.
- **Judging when the cross disappears is subjective:** use the same observer and the same cross each time, or use a light sensor/data logger.
- **Temperature changes:** the reactions are exothermic; carry out all runs at the same room temperature, or use a water bath.
- **Magnesium oxide coating:** clean the magnesium with emery paper so the surface is the same each time.
- **Random error:** repeat each concentration and calculate a mean.

Safety Considerations

- **Sulfur dioxide:** produced in Method B — it is toxic and may trigger asthma. Work in a well-ventilated room and dispose of the mixture promptly (pour into a sodium carbonate solution if instructed).

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- **Hydrogen:** flammable — keep away from naked flames.
- **Acids:** irritant; wear eye protection.

RP6 — Chromatography

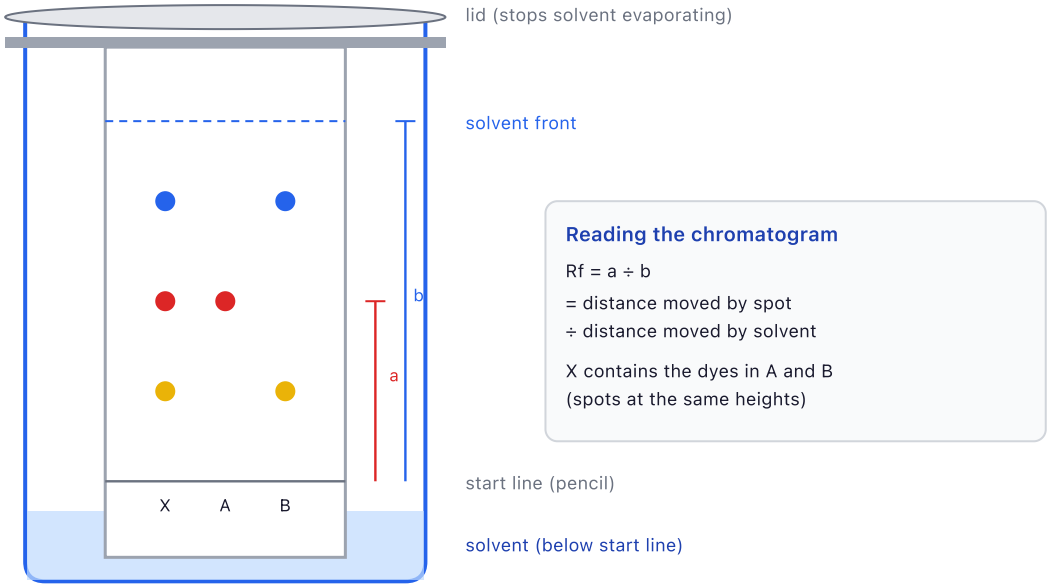
> **Aim:** Investigate how paper chromatography can be used to separate and tell the difference between coloured substances, and calculate R_f values (e.g. separating the dyes in inks or food colourings).

Apparatus

- Chromatography paper
- Beaker (250 cm^3) with a lid (e.g. a watch glass)
- Glass rod or wooden splint and paper clip (to hang the paper)
- Pencil and ruler
- Capillary tubes (or fine dropper) for spotting
- Samples: food colourings or inks, including known reference colours
- Solvent: water (or ethanol for inks that do not dissolve in water)

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RP6: Paper Chromatography



Reading the chromatogram

Rf = a ÷ b

= distance moved by spot
÷ distance moved by solvent

X contains the dyes in A and B
(spots at the same heights)

Method

1. Draw a horizontal **start line in pencil** about 2 cm from the bottom of the chromatography paper.
2. Use a capillary tube to put a small, concentrated spot of each sample on the line, spaced about 1.5 cm apart. Label each spot in pencil below the line.
3. Pour solvent (water) into the beaker to a depth of about 1 cm — the solvent level must be **below the start line**.
4. Hang the paper from a glass rod so the bottom edge dips into the solvent without the paper touching the sides of the beaker. Cover with a lid.
5. Leave the solvent to move up the paper. When it is about 1 cm from the top, remove the paper and immediately mark the **solvent front** in pencil.
6. Allow the chromatogram to dry.
7. Measure the distance from the start line to the centre of each spot, and from the start line to the solvent front. Calculate the R_f value of each spot.

Key Measurements and Calculations

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- **R_f = distance moved by substance ÷ distance moved by solvent** (always between 0 and 1, no units)
- **Worked example:** spot moves 3.6 cm, solvent moves 8.0 cm → R_f = 3.6 ÷ 8.0 = 0.45
- **Pure substance:** produces a single spot in all solvents.
- **Mixture:** separates into two or more spots.
- **Identification:** compare spots/R_f values with those of known substances run on the same paper with the same solvent.
- **Phases:** the paper is the **stationary phase**; the solvent is the **mobile phase**. Substances separate because they distribute differently between the phases — those more attracted to the solvent travel further.

> **Exam Tip:** The start line is drawn in pencil because graphite is insoluble in the solvent and will not move up the paper. Ink would dissolve and run, interfering with the results. Keep the solvent below the start line so the spots do not dissolve into the solvent in the beaker.

Sources of Error & Minimisation

- **Spots too large:** large spots spread and overlap — apply small spots, letting each dry before adding more.
- **Solvent evaporating:** cover the beaker with a lid so the atmosphere is saturated with solvent.
- **Paper touching the beaker sides:** causes the solvent to move unevenly — hang the paper centrally.
- **Not marking the solvent front immediately:** it evaporates and becomes invisible — mark it as soon as the paper is removed.
- **Insoluble substances:** stay on the start line; try a different solvent.

Safety Considerations

- **Ethanol (if used as solvent):** highly flammable — keep away from flames.
- **Inks/dyes:** may stain skin and clothing; some may be harmful — avoid contact and do not taste food colourings.

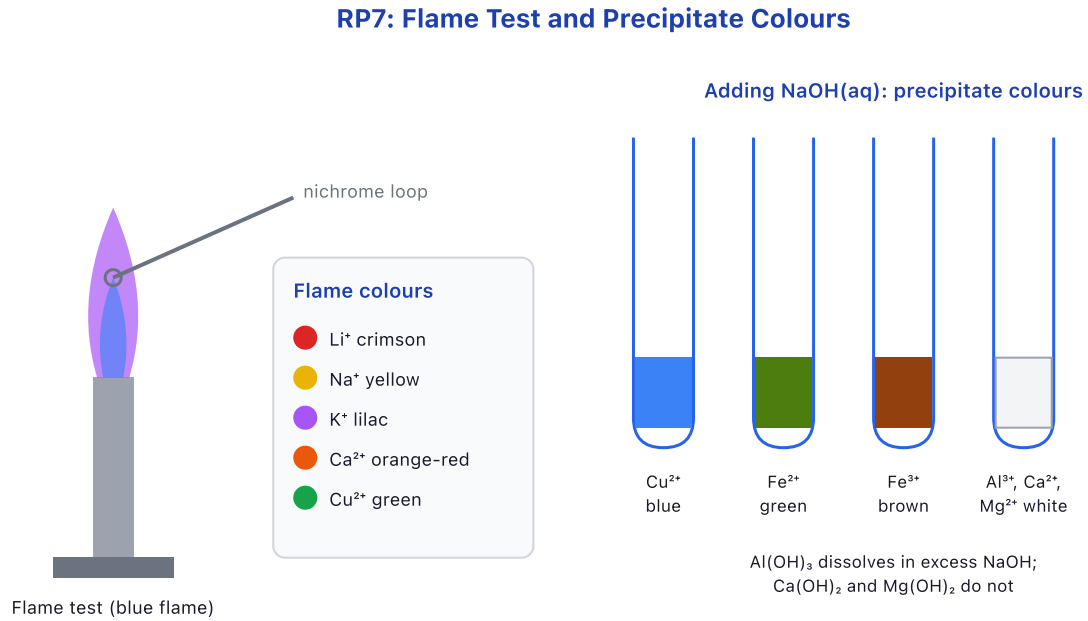
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RP7 — Identifying Ions (chemistry only)

Aim: Use chemical tests to identify the ions in unknown single ionic compounds, covering flame tests for metal ions, precipitation reactions with sodium hydroxide, and tests for carbonate, sulfate and halide ions.

Apparatus

- Test tubes and test-tube rack, dropping pipettes
- Nichrome wire loop (or wooden splints soaked in water), Bunsen burner
- Dilute hydrochloric acid (for cleaning the loop and for the carbonate and sulfate tests)
- Sodium hydroxide solution (0.4 mol/dm³)
- Dilute nitric acid and silver nitrate solution (halides)
- Barium chloride solution (sulfates)
- Limewater, delivery tube and bung (carbonates)
- Unknown solid compounds and their solutions
- Eye protection



Method — Flame Tests (metal ions)

1. Clean a nichrome wire loop by dipping it in dilute hydrochloric acid and holding it in a hot (blue) Bunsen flame until it gives no colour.
2. Dip the loop in dilute hydrochloric acid, then into the solid sample.
3. Hold the loop at the edge of the blue flame and observe the colour.
4. Clean the loop thoroughly between samples.

Method — Sodium Hydroxide Test (metal ions)

1. Put about 1 cm depth of the unknown solution in a test tube.
2. Add a few drops of dilute sodium hydroxide solution and note the colour of any precipitate.
3. If a white precipitate forms, add excess sodium hydroxide: if the precipitate dissolves, Al^{3+} is present. Use a flame test to distinguish Ca^{2+} (orange-red) from Mg^{2+} (no colour).

Method — Tests for Negative Ions

1. **Carbonate (CO_3^{2-}):** add dilute hydrochloric acid to the sample. Bubble any gas through limewater — carbon dioxide turns it milky.
2. **Halides (Cl^-, Br^-, I^-):** add dilute nitric acid, then a few drops of silver nitrate solution. Chloride gives a **white** precipitate, bromide a **cream** precipitate, iodide a **yellow** precipitate.
3. **Sulfate (SO_4^{2-}):** add dilute hydrochloric acid, then a few drops of barium chloride solution. A **white** precipitate (barium sulfate) shows sulfate ions.

Key Measurements and Calculations — Summary of Results

Ion	Test	Positive result
Li^+ / Na^+ / K^+ / Ca^{2+} / Cu^{2+}	Flame test	Crimson / yellow / lilac / orange-red / green
Cu^{2+}	NaOH(aq)	Blue precipitate
Fe^{2+}	NaOH(aq)	Green precipitate
Fe^{3+}	NaOH(aq)	Brown precipitate

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Al^{3+}	NaOH(aq)	White precipitate, dissolves in excess
Ca^{2+} / Mg^{2+}	NaOH(aq)	White precipitate, does not dissolve in excess
CO_3^{2-}	Dilute acid, then limewater	Fizzing; gas turns limewater milky
Cl^- / Br^- / I^-	HNO_3 then $AgNO_3$(aq)	White / cream / yellow precipitate
SO_4^{2-}	HCl then $BaCl_2$(aq)	White precipitate

- **Ionic equations (Higher):** $Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$; $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$; $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$

> **Exam Tip:** Acid is added first in the halide and sulfate tests to remove carbonate ions, which would also form a precipitate. Use nitric acid (not hydrochloric) in the halide test, because hydrochloric acid adds chloride ions; use hydrochloric acid (not sulfuric) in the sulfate test, because sulfuric acid adds sulfate ions.

Sources of Error & Minimisation

- **Contaminated wire loop:** clean in acid and flame between every test; sodium contamination gives a strong yellow that masks other colours.
- **Mixtures of ions:** some flame colours mask others — flame tests work best on single compounds; flame emission spectroscopy is more reliable for mixtures.
- **Adding too much sodium hydroxide at first:** add only a few drops initially, otherwise an Al^{3+} precipitate may dissolve before it is seen.
- **Subjective colour judgements:** compare with known samples and view against a white background.

Safety Considerations

- **Sodium hydroxide:** corrosive/irritant to eyes — wear eye protection.
- **Silver nitrate:** irritant and stains skin; dangerous to the environment.

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- **Barium chloride:** toxic if swallowed — handle with care and wash hands.
- **Bunsen burner:** tie back long hair; use a heatproof mat.
- **Unknown compounds:** treat as harmful; do not touch with fingers.

RP8 — Water Purification

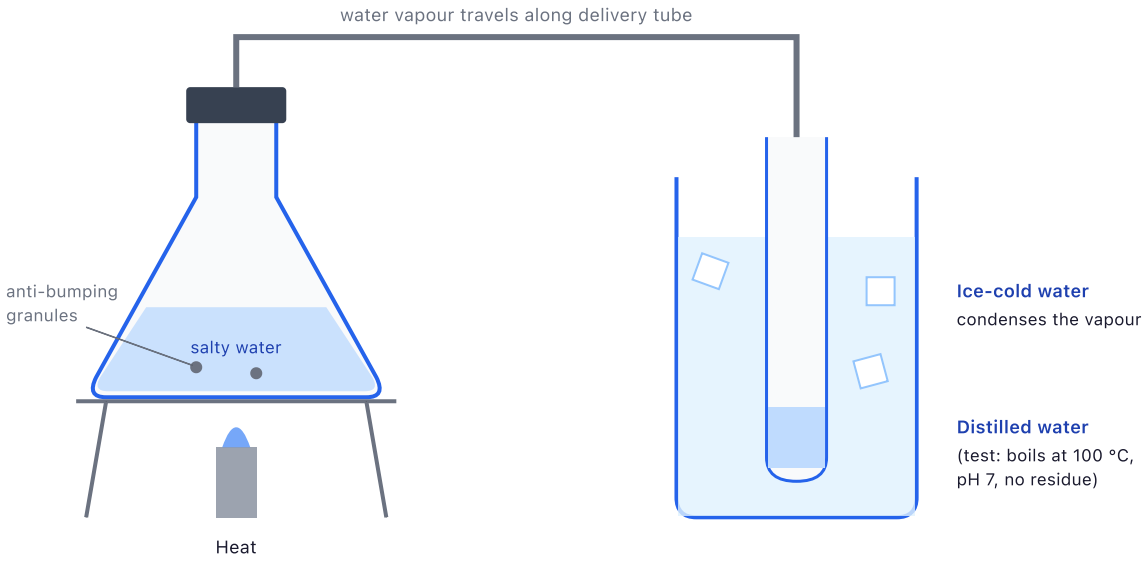
> **Aim:** Analyse and purify water samples from different sources, including measuring pH, measuring the mass of dissolved solids, and purifying a water sample by distillation.

Apparatus

- Water samples (e.g. tap water, sea water, rain water, pond water)
- Universal indicator paper or pH meter
- Evaporating basin, balance (±0.01 g), Bunsen burner, tripod and gauze (or water bath)
- Conical flask with bung and delivery tube
- Test tube in a beaker of ice-cold water (to condense the vapour), or a Liebig condenser
- Anti-bumping granules
- Thermometer
- Eye protection

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RP8: Simple Distillation of Salty Water



Method — Analysing Water Samples

1. **pH:** dip universal indicator paper into each sample (or use a calibrated pH meter) and compare the colour with the pH chart.
2. **Dissolved solids:** weigh a clean, dry evaporating basin. Add a measured volume (e.g. 25 cm^3) of the water sample and reweigh.
3. Heat the basin gently (over a water bath or Bunsen burner) until all the water has evaporated.
4. Allow to cool, then reweigh the basin. The increase in mass is the mass of dissolved solids.

Method — Purifying by Distillation

1. Pour about 50 cm^3 of the salty water sample into the conical flask and add a few anti-bumping granules.
2. Fit the bung and delivery tube, with the end of the tube inside a test tube standing in a beaker of ice-cold water.
3. Heat the flask so the water boils gently. Water vapour travels along the delivery tube and **condenses** in the cold test tube.
4. Collect a few cm^3 of distillate and then stop heating — **remove the delivery tube from the test tube first** to prevent suck-back.
5. Test the purity of the distilled water: check its pH (should be 7), evaporate a sample to check for no solid residue, or measure its boiling point (100 °C).

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Key Measurements and Calculations

- **Mass of dissolved solids** = mass of basin + residue − mass of empty basin
- **Concentration of dissolved solids (g/dm³)** = mass of solids (g) ÷ volume of sample (dm³)
- **Worked example:** 0.85 g of solid from 25 cm³ of sea water → 0.85 ÷ 0.025 = 34 g/dm³
- **Pure water:** pH 7, boils at exactly 100 °C, leaves no residue on evaporation.
- **Potable water** is safe to drink but is *not* pure — it contains low levels of dissolved salts and microbes.

> **Exam Tip:** Distillation is used to desalinate sea water on a small scale, but it needs large amounts of energy, so it is expensive. In the UK, potable water is usually produced from fresh water by filtration followed by sterilisation with chlorine, ozone or ultraviolet light.

Sources of Error & Minimisation

- **Vapour escaping:** make sure the bung fits tightly and the test tube is well cooled with ice.
- **Salty water bumping or splashing into the delivery tube:** use anti-bumping granules and heat gently.
- **Residue spitting out of the basin:** heat gently, or use a water bath, especially as the last water evaporates.
- **Small masses of residue:** use a balance with a resolution of 0.01 g and a larger volume of sample to reduce percentage uncertainty.
- **pH paper resolution:** a pH meter gives a more precise value than universal indicator paper.

Safety Considerations

- **Hot apparatus and steam:** steam can scald — do not touch the delivery tube or flask while hot.
- **Suck-back:** remove the delivery tube from the collecting tube before stopping heating, or cold water may be sucked back and crack the hot flask.
- **Never drink** any sample in the laboratory, including the distilled water.
- **Pond water:** may contain microorganisms — wash hands after handling.

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